

Noncommutative Transport Topology in the Rubik's Cube Representation

Sector Non-Invariance, Direct Support, and Transport Channels

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Abstract

Problem. A spectral sectorization need not be preserved by the operators whose transport it is used to study. The resulting off-diagonal blocks define a direct support graph. However, the relations among projector non-invariance, physical-block support, and the computed graph structure must be stated without assuming that the sector projectors lie in a group commutant.

Approach. We first establish an exact finite-dimensional identity relating off-diagonal transport to projector non-invariance, together with a separate block-locality theorem. The computational layer registers the Rubik sectors, direct blocks, support localizer, and finite EP algebra without promoting graph paths to matrix compositions.

Results. For a finite-dimensional Hilbert space, a complete orthogonal sectorization $I = \sum_{\alpha} Q_{\alpha}$, and a unitary operator U ,

$$\sum_{\beta \neq \alpha} \|Q_{\beta} U Q_{\alpha}\|_F^2 = \|(I - Q_{\alpha}) U Q_{\alpha}\|_F^2 = \frac{1}{2} \| [U, Q_{\alpha}] \|_F^2.$$

Thus off-diagonal transport is exactly the Frobenius mass of sector non-invariance. A separate block-locality theorem shows that direct transport vanishes between sectors with disjoint support whenever both the transport family and the sector projectors preserve the declared physical blocks.

Computational case study. In the declared complex128 Rubik realization, the quarter-turn and half-turn averages register nine numerical joint-spectral sectors. The 18 face-turn matrices produce a symmetric ten-edge direct graph. Nine certified edges have shared thresholded noncommutative block support; the remaining S8–S9 edge is carried by the numerically commuting corner-permutation block. The block-level noncommutative-support test serves as a localizer rather than a sufficient transport criterion: it selects 15 candidate pairs, of which six are nonedges. The computed EP averaging algebra $A_{EP} \cong M_2(\mathbb{C})^4 \oplus M_1(\mathbb{C})^4$ supplies a finite algebraic description of the dominant noncommutative block, but the direct matrices $Q_{\beta} \rho(g) Q_{\alpha}$ remain the transport certificate.

Boundary. Exact finite-dimensional statements, numerical Rubik registration, and mechanism labels are kept separate. Generator-family field censuses, negative controls, and exhaustive auxiliary tables are reported in supplementary material rather than presented as theorem claims. Graph paths are not interpreted as matrix compositions in this paper.

Notation

Symbol	Meaning
V	finite-dimensional Hilbert space; $\mathbb{C}^{228}$ in the Rubik case
$\rho(g)$	unitary transport operator associated with g
P_λ^A	spectral projector of the averaging operator A
Q_α	projector onto a QH joint-spectral sector
$T_{\beta\alpha}(g)$	$Q_\beta \rho(g) Q_\alpha$, from source α to target β
$K_{\beta\alpha}^S$	$\max_{g \in S} \ T_{\beta\alpha}(g)\ _F$
Π_b	projector onto physical block $b \in \{\text{cp}, \text{ep}, \text{co}, \text{eo}\}$
$\text{BlockSupp}(Q_\alpha)$	$\{b : \Pi_b Q_\alpha \neq 0\}$
$\text{Supp}_{\text{nc}}(\alpha)$	thresholded block-level QT noncommutative support
A_{EP}	algebra generated by the per-axis QT averages on the EP block

The direction convention is fixed throughout:

$$Q_\beta \rho(g) Q_\alpha \neq 0 \iff \alpha \longrightarrow \beta.$$

1 Introduction

Let $V = \bigoplus_\alpha V_\alpha$ be an orthogonal sectorization with projectors Q_α . Given a unitary family $\{\rho(g)\}$, the basic direct transport object is not an abstract edge but the matrix block

$$T_{\beta\alpha}(g) = Q_\beta \rho(g) Q_\alpha.$$

Its nonzero pattern depends on both the sectorization and the declared operator family. It cannot be inferred from sector dimensions, eigenvalue proximity, or block overlap alone. It also does not require the projectors to commute with the transport action. In fact, nontrivial off-diagonal transport requires precisely the opposite: at least one sector projector must fail to commute with a declared transport operator.

Throughout this paper, **transport topology** means the finite direct-support pattern induced by the declared operator blocks, not a topology in the point-set sense.

This paper has two layers.

1. The exact layer defines direct transport for an arbitrary complete orthogonal sectorization and proves the Transport–Non-Invariance Identity alongside a block-locality theorem.
2. The computational layer registers a nine-sector Rubik realization and reports its direct graph, a thresholded noncommutative-support localizer, the corner-permutation exception, and the corresponding EP algebra census.

The claim boundary is deliberate. Machine-zero commutators do not prove exact commutation. The nine registered Rubik projectors are computational objects unless an exact joint-spectrum certificate is supplied. The Type I/II labels classify already-certified direct edges; they are not universal sufficient conditions for transport. A separate analysis [3] studies whether paths in a support graph survive as projected matrix products.

The main contribution therefore has two parts: direct off-diagonal mass is exactly sector non-invariance, while the Rubik algebraic diagnostics describe how that mass is distributed in one finite registered system. The certified direct blocks and edge labels can serve as inputs to routed-composition audits, but do not themselves promote a graph path to a nonzero product.

2 Spectral and Joint-Spectral Sectors

2.1 Rubik representation and averaging families

The registered Rubik representation is block diagonal on

$$V = V_{\text{cp}} \oplus V_{\text{ep}} \oplus V_{\text{co}} \oplus V_{\text{eo}},$$

with dimensions $64 + 144 + 8 + 12 = 228$, respectively. The generator set $S = S^{-1}$ contains the 18 standard face turns. Every $\rho(g)$ preserves these four physical blocks.

Separate the 12 quarter turns and six half turns and define their uniform averages QT_{all} and HT_{all} . By construction,

$$A_{18} = \frac{1}{18} \sum_{g \in S} \rho(g) = \frac{2}{3} \text{QT}_{\text{all}} + \frac{1}{3} \text{HT}_{\text{all}}.$$

The A_{18} -spectral projectors

$$P_{\lambda}^A = \mathbf{1}_{\{\lambda\}}(A_{18})$$

and the QH joint-spectral projectors discussed below are different objects. The former yield six coarse averaging layers in the canonical computation. The latter resolve those layers into nine sectors and are the projectors used for transport.

2.2 Exact commuting model

Suppose Q and H are commuting Hermitian operators with spectral projectors E_q^Q and E_h^H . For every nonzero joint eigenspace define

$$Q_{\alpha} = E_{q_{\alpha}}^Q E_{h_{\alpha}}^H, \quad V_{\alpha}^{QH} = \text{im } Q_{\alpha}.$$

The Q_{α} form a complete orthogonal sectorization. For an affine combination $A = aQ + bH$, its spectral projectors are collision quotients:

$$P_{\lambda}^A = \sum_{\alpha: aq_{\alpha} + bh_{\alpha} = \lambda} Q_{\alpha}.$$

This mathematical statement is exact under the stated commuting-Hermitian hypotheses. It does not assert that a numerically registered pair commutes exactly.

2.3 Computational Rubik registration

In the declared complex128 realization, the three relevant commutator residuals are

$$\begin{aligned} \| [A_{18}, \text{QT}_{\text{all}}] \|_F &= 1.58 \times 10^{-16}, \\ \| [A_{18}, \text{HT}_{\text{all}}] \|_F &= 1.01 \times 10^{-17}, \\ \| [\text{QT}_{\text{all}}, \text{HT}_{\text{all}}] \|_F &= 3.77 \times 10^{-16}. \end{aligned}$$

Simultaneous numerical registration produces nine QH joint-spectral sectors. Their projector completeness residual is 1.34×10^{-14} and the maximum cross-sector product norm is 3.31×10^{-15} .

Sector	Dim	λ_A	λ_{QT}	λ_{HT}	Physical support
S1	20	1	1	1	cp(8)+ep(12)
S2	2	8/9	5/6	1	eo(2)
S3	39	7/9	5/6	2/3	ep(36)+eo(3)
S4	26	2/3	1/2	1	ep(24)+co(2)
S5	1	5/9	1/3	1	eo(1)
S6	39	5/9	1/2	2/3	ep(36)+eo(3)
S7	66	5/9	2/3	1/3	cp(24)+ep(36)+co(3)+eo(3)
S8	8	1/3	0	1	cp(8)
S9	27	1/3	1/3	1/3	cp(24)+co(3)

These values are a numerical registration against displayed rational labels. The exact identity $A_{18} = (2/3)QT_{\text{all}} + (1/3)HT_{\text{all}}$ does not by itself promote the registered joint spectrum to exact arithmetic.

3 Direct Transport and Sector Non-Invariance

3.1 Definitions

Definition 3.1 (Sectorized transport datum).

Let V be a finite-dimensional Hilbert space, let $\rho : G \rightarrow U(V)$ be unitary, let $S \subseteq G$, and let

$$I = \sum_{\alpha \in \mathcal{I}} Q_{\alpha}, \quad Q_{\alpha} Q_{\beta} = \delta_{\alpha\beta} Q_{\alpha}$$

be a complete orthogonal sectorization. No assumption $Q_{\alpha} \in \text{End}_G(V)$ is made.

Definition 3.2 (Direct transport).

For source α , target β , and $g \in S$, define

$$T_{\beta\alpha}(g) = Q_{\beta} \rho(g) Q_{\alpha}, \quad K_{\beta\alpha}^S = \max_{g \in S} \|T_{\beta\alpha}(g)\|_F.$$

Exact direct support is $\mathbf{1}[K_{\beta\alpha}^S > 0]$. A numerical graph replaces zero by a declared threshold τ_K . If $S = S^{-1}$, then unitarity and Frobenius-norm invariance give $K_{\beta\alpha}^S = K_{\alpha\beta}^S$.

3.2 Transport–Non-Invariance Identity

Proposition 3.3 (Off-Diagonal Transport–Non-Invariance Identity).

For every unitary U and every sector projector Q_{α} ,

$$\sum_{\beta \neq \alpha} \|Q_{\beta} U Q_{\alpha}\|_F^2 = \|(I - Q_{\alpha}) U Q_{\alpha}\|_F^2 = \|[U, Q_{\alpha}] Q_{\alpha}\|_F^2 = \frac{1}{2} \|[U, Q_{\alpha}]\|_F^2.$$

Proof. Relative to $V = V_{\alpha} \oplus V_{\alpha}^{\perp}$, write

$$U = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad Q_{\alpha} = \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix}.$$

Then

$$[U, Q_{\alpha}] = \begin{pmatrix} 0 & -B \\ C & 0 \end{pmatrix}, \quad [U, Q_{\alpha}] Q_{\alpha} = \begin{pmatrix} 0 & 0 \\ C & 0 \end{pmatrix}.$$

Orthogonality of $I - Q_\alpha = \sum_{\beta \neq \alpha} Q_\beta$ gives

$$\|C\|_F^2 = \|(I - Q_\alpha)UQ_\alpha\|_F^2 = \sum_{\beta \neq \alpha} \|Q_\beta UQ_\alpha\|_F^2.$$

Unitarity implies $A^*A + C^*C = I$ and $AA^* + BB^* = I$. Taking traces yields $\|B\|_F^2 = \|C\|_F^2$. Therefore, $\|[U, Q_\alpha]\|_F^2 = \|B\|_F^2 + \|C\|_F^2 = 2\|C\|_F^2$.

Corollary 3.4 (Sector invariance).

If $S = S^{-1}$ generates G , then the following are equivalent:

1. Q_α commutes with $\rho(G)$;
2. V_α is G -invariant;
3. $Q_\beta \rho(g) Q_\alpha = 0$ for all $g \in S$ and $\beta \neq \alpha$;
4. the outgoing off-diagonal mass in Proposition 3.3 is zero for every $g \in S$.

If $Q_\beta \rho(g) Q_\alpha \neq 0$ with $\alpha \neq \beta$, then neither endpoint projector commutes with $\rho(g)$. Indeed, commutation of Q_α would give

$$Q_\beta \rho(g) Q_\alpha = Q_\beta Q_\alpha \rho(g) = 0,$$

and commutation of Q_β would give

$$Q_\beta \rho(g) Q_\alpha = \rho(g) Q_\beta Q_\alpha = 0.$$

3.3 Block locality

Theorem 3.5 (Direct Block-Locality).

Let $I = \sum_b \Pi_b$ be an orthogonal physical-block decomposition. Suppose every $\rho(g)$ and every Q_α commutes with every Π_b . If

$$\text{BlockSupp}(Q_\alpha) \cap \text{BlockSupp}(Q_\beta) = \emptyset,$$

then $Q_\beta \rho(g) Q_\alpha = 0$ for every g .

Proof. Insert $I = \sum_b \Pi_b$ and use block preservation:

$$Q_\beta \rho(g) Q_\alpha = \sum_b Q_\beta \Pi_b \rho(g) \Pi_b Q_\alpha.$$

For each b , disjoint block support makes at least one endpoint factor zero.

This theorem is only a zero criterion. Shared physical-block support does not imply nonzero direct transport.

4 Canonical Rubik Direct Graph

4.1 Numerical policy

The canonical direct graph uses:

Field	Value
representation	complex128, dimension 228
operator family	18 face-turn matrices
sectorization	nine registered QH projectors
block norm	Frobenius norm
direct-edge threshold	$\tau_K = 0.05$
noncommutativity threshold	$\tau_{\text{nc}} = 10^{-10}$

The smallest retained off-diagonal edge has norm 0.471, while unretained pairs are below 10^{-14} in the canonical audit. Thus the ten-edge census is unchanged throughout the threshold interval from 10^{-10} to 0.1.

4.2 Direct edge census

The complete K matrix is provided with the supplementary data. Its off-diagonal support consists of ten direct edges.

An unordered edge is retained when $K_{\beta\alpha}^S > \tau_K$. Generator-level block counts and witness multiplicities are reported in the supplementary data.

Edge	K	Shared block	Registered label
S2–S5	0.47	eo	Type I
S2–S6	0.58	eo	Type I
S3–S6	2.55	ep, eo	Type I
S3–S7	3.61	ep, eo	Type I
S4–S6	3.46	ep	Type I
S4–S9	1.00	co	Type I
S5–S6	0.82	eo	Type I
S6–S7	3.61	ep, eo	Type I
S7–S9	4.06	cp, co	Type I
S8–S9	2.83	cp	Type II

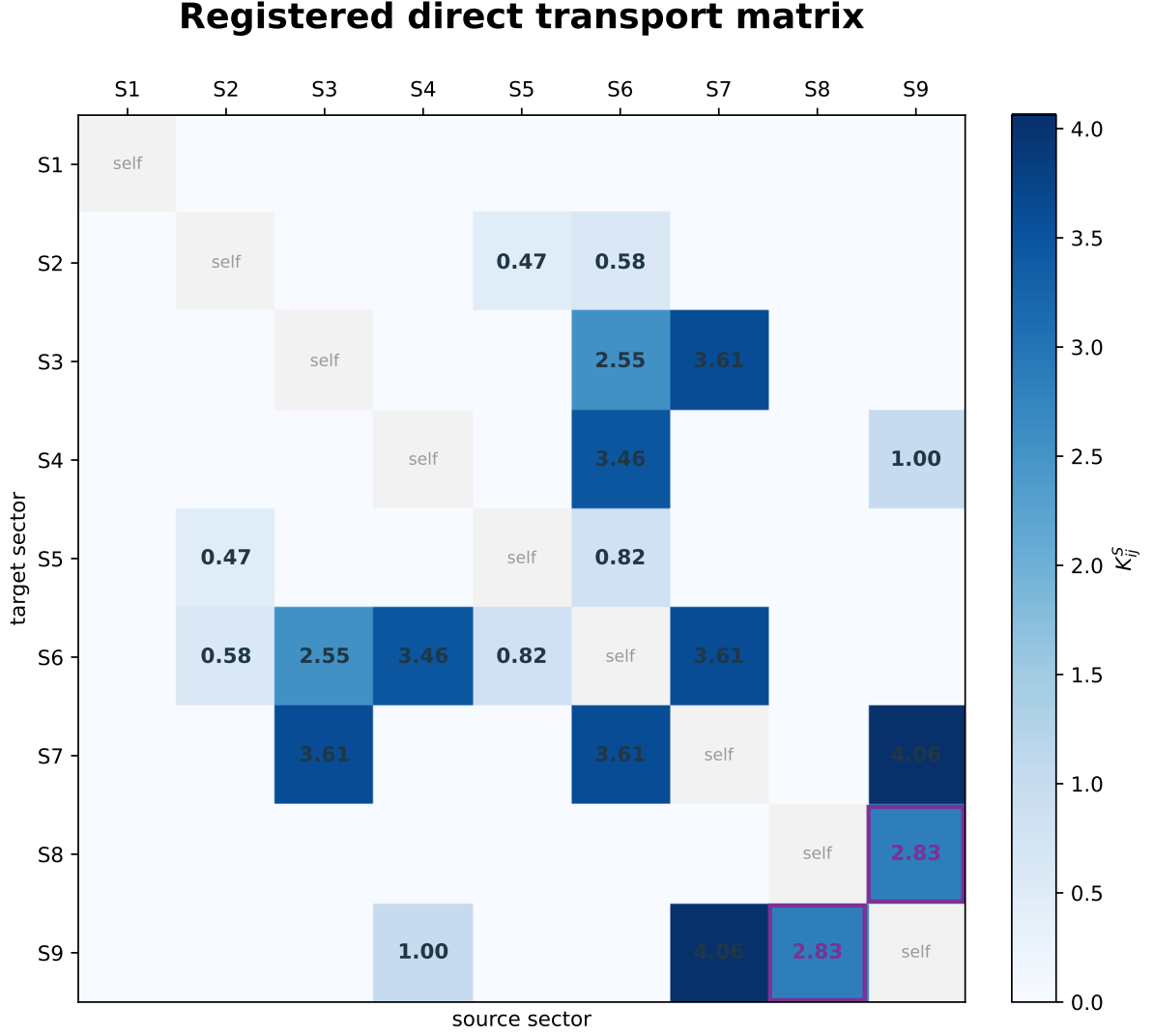


Figure 1. The registered 9×9 direct transport matrix. Off-diagonal cells record the ten aggregate unordered edges; the purple border marks the S8–S9 Type II labelled edge, and diagonal self-blocks are omitted from the color scale. Labels are assigned after direct block certification.

The graph is symmetric to $\max |K - K^T| = 1.11 \times 10^{-16}$. Its degree sequence is

$$(0, 2, 2, 2, 2, 5, 3, 1, 3).$$

Consequently, S6 is the unique degree-five hub. S1 is numerically isolated: $K_{1\beta} < 10^{-14}$ for $\beta \neq 1$. By Corollary 3.4, this is a numerical sector-invariance certificate, not an exact G -invariance theorem.

Every edge is block-local, as required by Theorem 3.5. The converse fails: many sectors share a physical block but have zero direct transport.

4.3 Noncommutative support is a localizer

Let QT^a , $a \in \{0, 1, 2\}$, be the three per-axis quarter-turn averages. For the declared threshold define

$$\text{Supp}_{\text{nc}}(\alpha) = \left\{ b : \Pi_b Q_\alpha \neq 0, \max_{0 \leq a < c \leq 2} \|\Pi_b[\text{QT}^a, \text{QT}^c]\Pi_b\|_F > \tau_{\text{nc}} \right\}.$$

In the canonical audit, all three axis pairs $(0, 1)$, $(0, 2)$, $(1, 2)$ give the same block norms to the displayed precision:

Block	Norm	Ratio to total norm
cp	$< 10^{-14}$	0% to reported precision
ep	2.7386	93.9%
co	0.6124	21.0%
eo	0.7906	27.1%

The ratios are not additive; Frobenius masses add in quadrature. The support sets are

Sector	Supp_{nc}
S1	$\emptyset$
S2	$\{\text{eo}\}$
S3	$\{\text{ep}, \text{eo}\}$
S4	$\{\text{ep}, \text{co}\}$
S5	$\{\text{eo}\}$
S6	$\{\text{ep}, \text{eo}\}$
S7	$\{\text{ep}, \text{co}, \text{eo}\}$
S8	$\emptyset$
S9	$\{\text{co}\}$

All nine non-CP direct edges have nonempty support intersection. However, such an intersection occurs for 15 unordered pairs in total. The six false-positive candidates are

$$\{\text{S2} - -\text{S3}, \text{S2} - -\text{S7}, \text{S3} - -\text{S4}, \text{S3} - -\text{S5}, \text{S4} - -\text{S7}, \text{S5} - -\text{S7}\}.$$

By definition of the registered Type I label, every Type I edge has shared block-level Supp_{nc} . The nontrivial empirical statement is instead

$$15 \text{ overlap candidates} = 9 \text{ certified Type I edges} + 6 \text{ nonedges}.$$

Thus the support is a candidate localizer and mechanism descriptor, not a transport certificate. Only $Q_\beta \rho(g) Q_\alpha$ or the equivalent leakage mass certifies a direct edge.

4.4 Type I and Type II are registered edge labels

The terminology is defined after direct-edge certification:

- **Type I:** a certified direct edge whose endpoints share thresholded noncommutative block support. There are nine such edges.
- **Type II:** a certified direct edge without such an intersection, carried here by a shared block whose per-axis QT commutator is below threshold. The unique instance is S8–S9 on CP.

This is an exhaustive finite classification of the ten registered edges, not a universal dichotomy for arbitrary representations.

The S8–S9 channel illustrates why averaged commutativity and generator commutativity must be separated. On CP,

$$\max_{0 \leq a < c \leq 2} \|\Pi_{\text{cp}}[\text{QT}^a, \text{QT}^c]\Pi_{\text{cp}}\|_F < 10^{-14},$$

yet the individual CP permutation matrices need not preserve either QH eigenspace. The directly evaluated block $Q_9 \rho(g) Q_8$ is nonzero. Thus a commuting averaging family can define sectors that are mixed by its unaveraged generators.

4.5 EP algebra census

The EP block carries most of the per-axis QT commutator norm. The three registered per-axis QT restrictions are Hermitian and generate a numerically closed 20-dimensional unital $*$ -algebra with an eight-dimensional center. Finite-dimensional unital $*$ -subalgebras of a matrix algebra are finite-dimensional C^* -algebras and hence semisimple [9]. The computational audit reports generator Hermiticity, identity-in-algebra, multiplication-closure, and $*$ -closure residuals of 0, 6.989×10^{-15} , 2.681×10^{-15} , and 1.416×10^{-14} , respectively. The finite numerical Artin–Wedderburn census is

$$A_{\text{EP}} \cong M_2(\mathbb{C})^4 \oplus M_1(\mathbb{C})^4.$$

Each of the four M_2 simple components occurs with representation multiplicity 12 and therefore occupies a 24-dimensional central component. Similarly, the four M_1 components each have multiplicity 12 and occupy 12-dimensional components:

$$4(2 \cdot 12) + 4(1 \cdot 12) = 144.$$

Three of the four registered 24-dimensional components have nonzero family-level commutator norm; the fourth and all four 12-dimensional components are commuting to the declared tolerance. This algebra localizes the dominant EP noncommutativity and provides coordinates for the finite Rubik analysis.

It does not prove the direct graph, force a unique hub, or identify components of the full G -commutant. The graph and hub degree are computed from the transport blocks. Detailed center, Killing-form, and double-commutant tables are reported separately and are not used in the proof of the graph census.

4.6 Graph/composition boundary

Some length-two support paths have endpoints on disjoint physical blocks. Such graph paths do not imply nonzero projected matrix products. For five canonical pairs, a separate composition audit reports machine-zero ordered products with image–kernel and block-support explanations [3]. These graph-only pairs are not composition morphisms, and the composition claims are not used to establish the direct graph.

5 QH Refinement Boundary

Proposition 5.1 (Atoms of the declared QH joint spectral resolution).

If QT_{all} and HT_{all} are Hermitian and commute exactly, then the Q_α are the primitive spectral idempotents of the represented commutative algebra

$$\mathcal{B}_{\text{QH}} = \text{alg}(\text{QT}_{\text{all}}, \text{HT}_{\text{all}}).$$

Equivalently, no spectral projector of an element of $\mathcal{B}_{\text{QH}}$ further splits a range $Q_\alpha V$.

Proof. *Commuting Hermitian operators admit a simultaneous orthogonal spectral resolution. Every element of the finite-dimensional commutative algebra is a polynomial in the declared generators and acts by a scalar on each common joint eigenspace. Its spectral projectors are sums of the primitive joint spectral idempotents Q_α .*

This statement concerns spectral idempotents within the represented commutative algebra. It does not imply that $Q_\alpha V$ is a minimal invariant subspace as an abstract module: because the algebra acts on this space by scalars, every subspace is invariant. Nor does it claim that the nine registered Rubik sectors constitute the finest orthogonal sectorization in $\text{End}(V)$.

The per-axis averages cannot all be adjoined to the same commuting spectral family: the registered norms satisfy

$$\|[\text{QT}^0, \text{QT}^1]\|_F = \|[\text{QT}^0, \text{QT}^2]\|_F = \|[\text{QT}^1, \text{QT}^2]\|_F = 2.9155.$$

This is an obstruction to simultaneous diagonalization of those particular operators. It is not a theorem excluding every larger commuting extension of $\mathcal{B}_{\text{QH}}$.

6 Claim Status and Boundary

The table uses the four claim levels. Theorem includes exact lemmas, propositions, and corollaries under their stated hypotheses.

Claim	Status
Transport–Non-Invariance Identity	Theorem
Sector-invariance equivalence	Theorem
Direct block-locality under the stated block-preservation hypotheses	Theorem
Primitive spectral idempotents of $\mathcal{B}_{\text{QH}}$ under the stated hypotheses	Theorem
Nine QH sectors registered against the displayed rational coordinates	Computational Certificate
Ten-edge K graph and degree sequence	Computational Certificate
Type I/II finite edge labels	Computational Certificate
A_{EP} Wedderburn census	Computational Certificate
S1 numerical G -invariance candidate	Computational Observation

The failure boundary is separate from the claim-status table:

- block-level Supp_{nc} does not determine transport; the registered census contains six false positives;
- a graph path does not imply projected composition;
- global maximality of the QH sectors among commuting refinements is not claimed.

The following supplementary records extend the computational Rubik case study:

- full-precision K matrix and all zero-pair records;
- generator-family and spectral-field comparisons;
- S3 negative-control tables;
- EP center, Killing-form, and commutant diagnostics;
- auxiliary figures and earlier transport interpretations.

These records provide reproducibility data and development context; they do not support additional theorem claims.

Appendix A indexes the executable artifacts. All numerical support uses Frobenius norms and explicit thresholds. Machine-zero is reported as computational evidence, not exact vanishing.

7 Discussion

The exact Transport–Non-Invariance Identity is independent of Rubik structure. Whenever a unitary family is expressed relative to a complete orthogonal sectorization, the total outgoing off-diagonal mass from a sector equals one half of the projector-commutator mass. This identity gives a complete generator-level answer to whether a sector is invariant.

The Rubik realization provides a finite computational case study. Its QH projectors are registered from a numerically commuting averaging pair but are not preserved by the full face-turn action. The resulting transport graph is sparse and block-local. Per-axis noncommutativity localizes all nine non-CP edges but does not determine them. The CP edge shows that a commuting averaged block may still support generator transport. The EP algebra explains where most of the averaged noncommutativity resides without replacing direct matrix evaluation.

7.1 Promotion limits

Two promotion problems remain open. First, a general algebraic criterion for direct transport needs more than block-level noncommutative support. Second, a graph path requires a separate product audit before it can be interpreted as projected-composition reachability.

8 Conclusion

For a complete orthogonal sectorization and unitary transport operator,

$$\sum_{\beta \neq \alpha} \|Q_\beta U Q_\alpha\|_F^2 = \frac{1}{2} \|[U, Q_\alpha]\|_F^2.$$

The central identity equates off-diagonal transport mass with sector non-invariance. The registered Rubik case has ten computed direct edges. Nine have nonempty Supp_{nc} intersection, while one uses the commuting CP block. This intersection is part of the Type I label definition; the registered data show that it is not sufficient for an edge. The computed transport matrices therefore remain the certificate. This separates block locality, mechanism labels, and composition boundaries without assuming $Q_\alpha \in \text{End}_G(V)$.

9 Related Work and Novelty Boundary

Finite-group representations and matrix blocks provide the ambient language [10, 4]. Finite-dimensional C^* -algebra structure and Artin–Wedderburn theory supply the semisimple algebra background [9, 8]. Random-walk averages on finite groups supply the averaging-operator viewpoint [5].

The commuting spectral side is related to Bose–Mesner algebras and quotient constructions [1, 6, 7]. Those theories organize commuting averages and their eigenspaces. The direct blocks studied here are additional data: they measure a separate operator family relative to that spectral decomposition.

Related work studies the canonical averaging spectrum and projected composition obstructions [2, 3]. Extended finite censuses are available as optional supplementary data.

Appendix A Computational Artifacts

The following repository artifacts provide the reproducible support layer for the Rubik case study. The default directory is `experiments/paper2/`; paths in the table are relative to that directory.

Artifact	Role	Short path
A1	numerical QH sector registration and projector checks	<code>validation/primitive_sectors.py</code>
A2	direct K graph and degree census	<code>validation/transport_graph.py</code>
A3	family-level QT commutator localizer	<code>validation/supp_nc.py</code>
A4	finite EP unital $*$ -algebra census	<code>validation/ep_algebra.py</code>
A5	subgroup/full-action boundary audit	<code>validation/symmetry_and_transport_audit.py</code>
A6	numerical QT/HT joint-spectrum audit	<code>validation/joint_spectral_geometry.py</code>
A7	exact affine collision arithmetic for the declared coordinates	<code>validation/collision_geometry.py</code>
A8	finite generator-family comparison	<code>validation/generator_universality.py</code>
A9	frozen display data for the direct-transport heatmap	<code>results/direct_transport.json</code>

From the repository root, an executable artifact is run as `python experiments/paper2/<short path>`. A6–A8 are finite or conditional controls and do not support additional theorem claims.

All listed artifacts are available in the [RIME repository](#).

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